INTRODUCTION TO DATA SCIENCE · FINAL PRESENTATION

Diabetes Risk Prediction

Predicting diabetes from 21 health indicators using supervised machine learning

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SEMESTER · DATE

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OUTLINE

What we will cover today

- 01 · Problem statement
- 02 · Dataset overview
- 03 · Target & features
- 04 · Project workflow
- 05 · Data cleaning
- 06 · Exploratory data analysis
- 07 · Correlation analysis
- 08 · Train / test split
- 09 · Why machine learning?
- 10 · 4 candidate ML models
- 11 · Model comparison
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- 14 · Streamlit web app
- 15 · Limitations
- 16 · Conclusion & Q&A

01 · WHY THIS MATTERS

The diabetes problem

- Diabetes affects 1 in 11 adults worldwide (~537 million people).
- Early detection drastically reduces complications and costs.
- Many people are unaware they have prediabetes or diabetes.
- Doctors need low-cost, fast screening tools.
- Patient survey data is cheap to collect — no lab tests required.

GLOBAL IMPACT

537M

adults living with diabetes

Source: IDF Diabetes Atlas 2021

Dataset overview

- Source: CDC Behavioral Risk Factor Surveillance System (BRFSS), 2015.
- Collected through annual phone interviews across the United States.
- Cleaned version published on Kaggle by Alex Teboul.
- Pre-balanced 50 / 50 between diabetic and non-diabetic patients.
- Each row = one patient · each column = one health indicator.

DATASET AT A GLANCE	
Records	70,693 patients
Features	21 health indicators
Target	Diabetes_binary (0 / 1)
Balance	50 % / 50 %
Type	Tabular · structured
Task	Binary classification

Target and features

TARGET VARIABLE (y)

Diabetes_binary → 0 = No Diabetes · 1 = Diabetes / Prediabetes

21 FEATURES (X)

VITAL · CLINICAL

- HighBP — high blood pressure
- HighChol — high cholesterol
- CholCheck — checked in 5 years
- BMI — Body Mass Index
- Stroke — ever had a stroke
- HeartDiseaseorAttack
- DiffWalk — difficulty walking

LIFESTYLE

- Smoker
- PhysActivity
- Fruits
- Veggies
- HvyAlcoholConsump
- AnyHealthcare
- NoDocbcCost

SELF-REPORTED · DEMOGRAPHICS

- GenHlth — general health (1-5)
- MentHlth — bad-mood days
- PhysHlth — sick days
- Sex — 0 female / 1 male
- Age — group (1-13)
- Education — level (1-6)
- Income — bracket (1-8)

Project workflow

01 Data Collection

Load CSV

02 Data Cleaning

Duplicates · dtypes

03 EDA & Correlation

Stats · heatmap

04 Model Training

Random Forest

05 Model Evaluation

Accuracy · matrix

06 Insights & App

Top factors · GUI

Data cleaning — what we did

- Checked for missing values → found 0 ✓
- Removed duplicate rows → 1 636 dropped
- Final shape: 69 057 rows × 22 columns
- Converted 21 columns to int (everything except BMI which is float)
- Verified class balance still ~50 / 50

BEFORE → AFTER

Rows

70 693 → 69 057

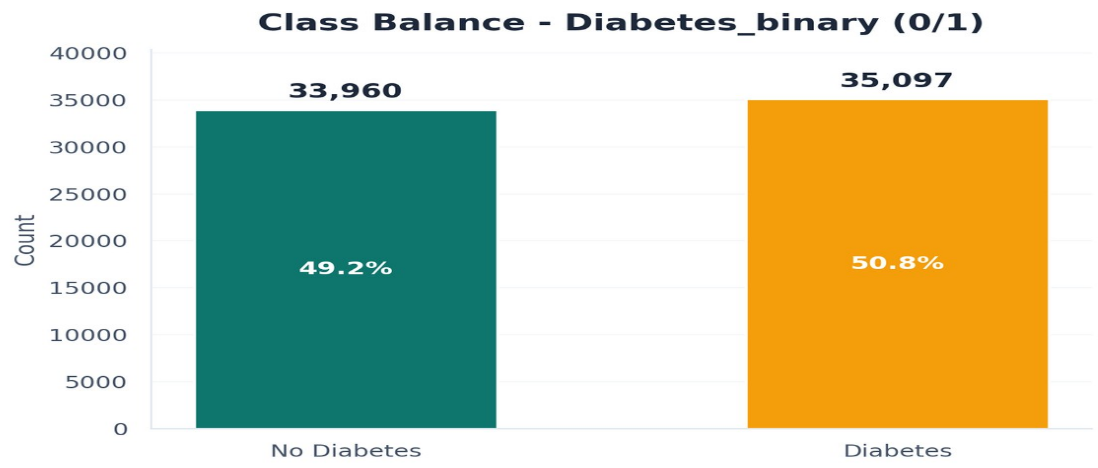
Duplicates

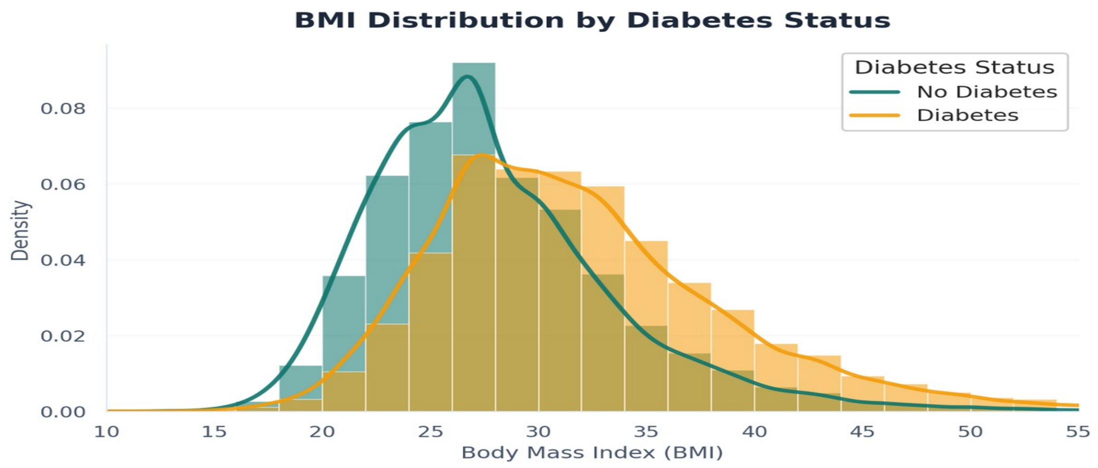
1 636 → 0

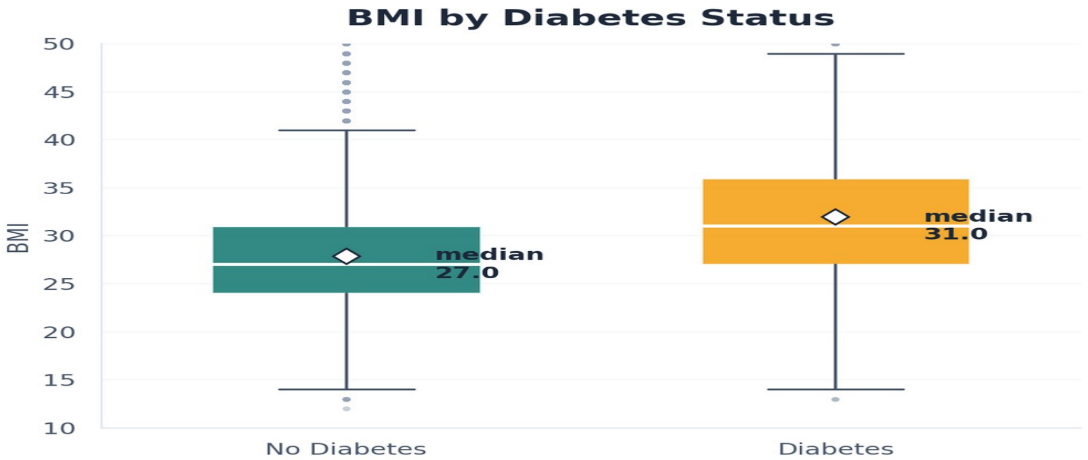
Missing values

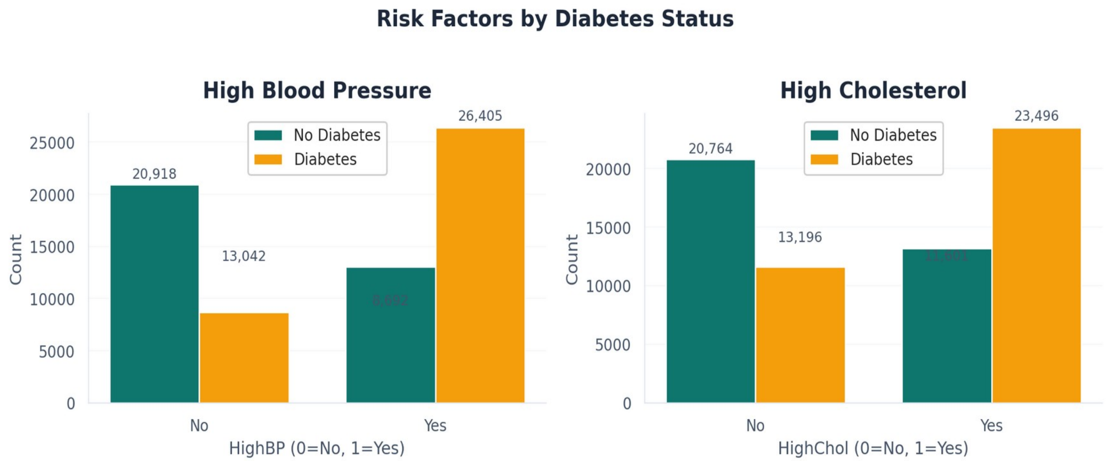
0 → 0 ✓ already clean

Exploratory data analysis

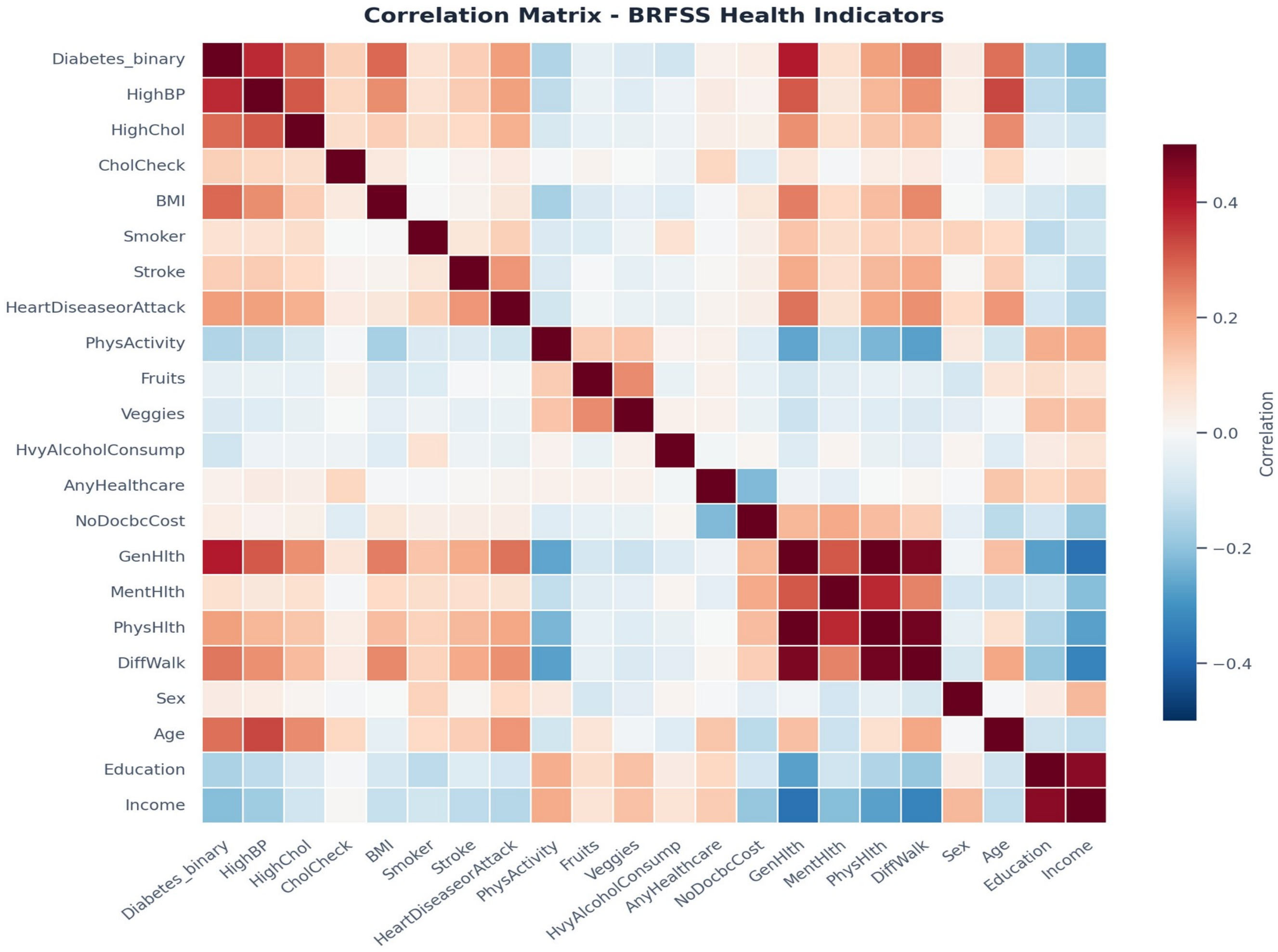




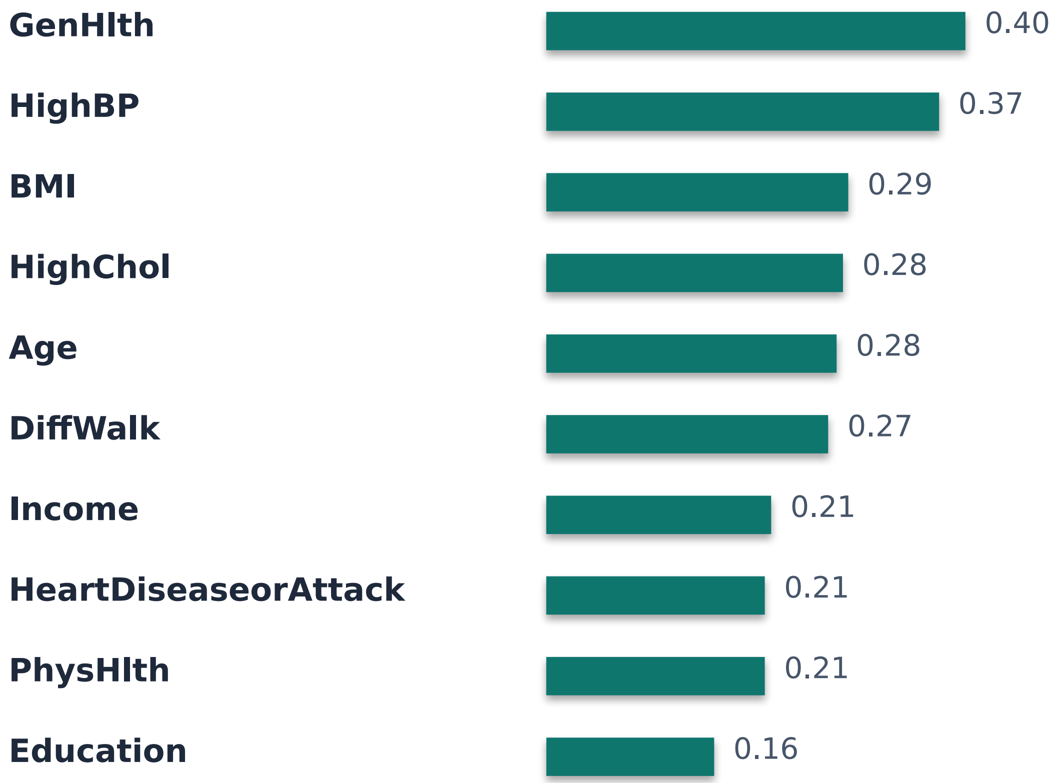




Which features matter most?



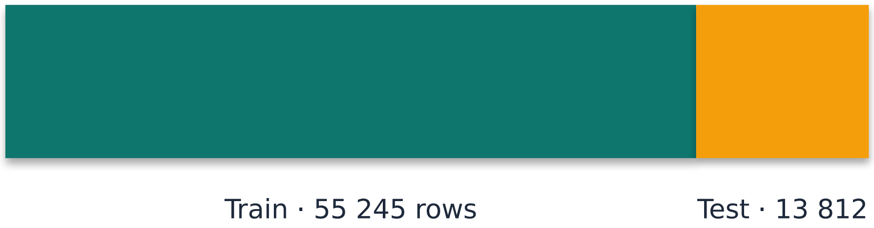
TOP 10 CORRELATED WITH DIABETES



Train / test split — done before everything else

- 80 % training data · 20 % test data
- Stratified by target → keeps 50 / 50 balance in both splits
- random_state = 42 → reproducible results
- Split BEFORE any preprocessing → prevents data leakage

DATA SPLIT



WHY SPLIT FIRST?

If we scaled, imputed, or sampled before splitting, information from the test set would leak into training. Accuracy would look great in our notebook but fail on new data.

Why supervised machine learning?

LEARNING TYPE

Supervised

The target column is labeled (0 / 1) — the model learns from known examples.

TASK

Classification

We predict a discrete class, not a number. Two classes → binary classification.

OUTPUT

Probability + label

Each prediction includes a confidence score, useful for risk-screening contexts.

4 supervised classifiers tested

LOGISTIC REGRESSION

Fits a probability line between the two classes

PROS Simple · interpretable

CONS Less accurate on non-linear patterns

DECISION TREE

Asks yes/no questions to split the data

PROS Easy to visualise

CONS Prone to overfitting

K-NEAREST NEIGHBORS

Predicts class by voting among nearest neighbours

PROS Intuitive

CONS Slow on large data · needs scaling

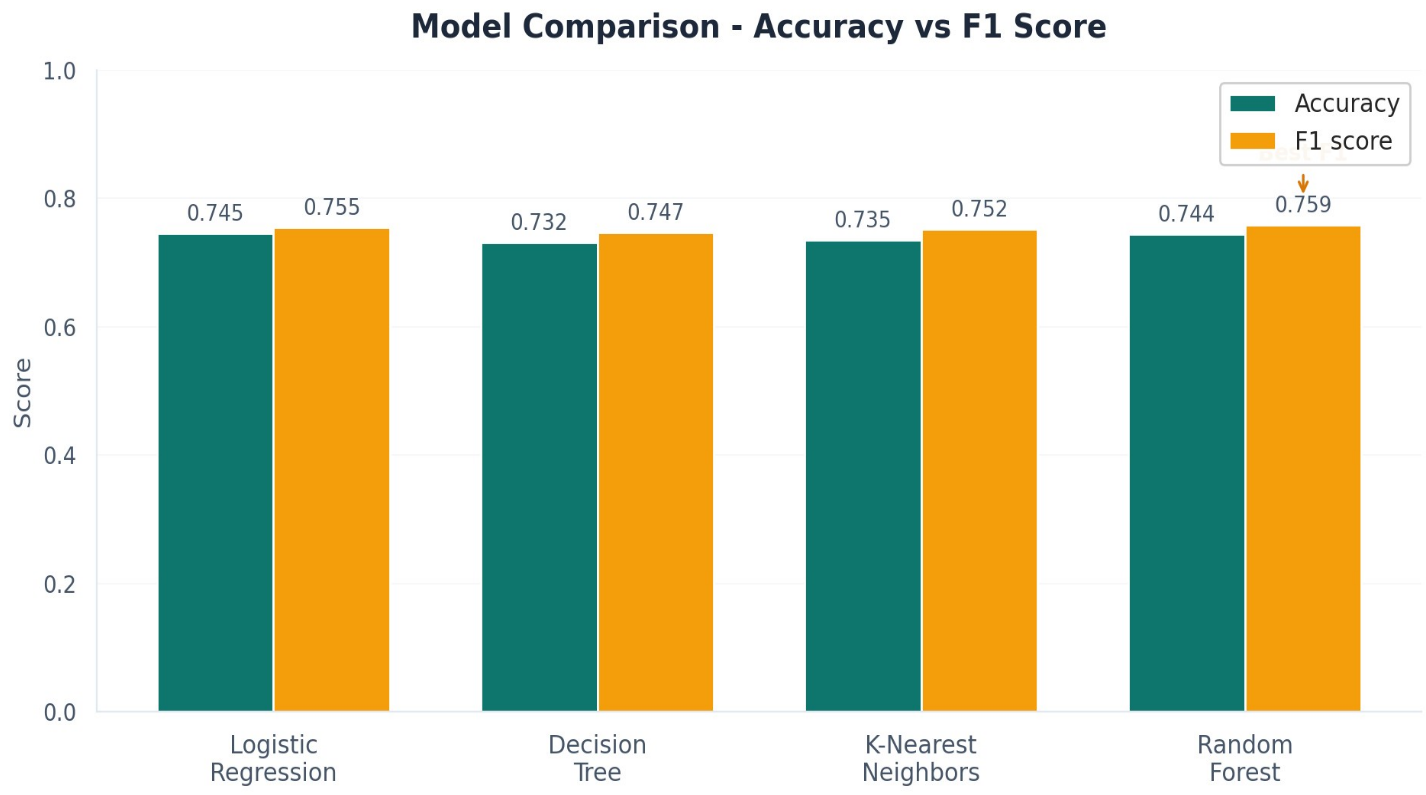
RANDOM FOREST

Combines many decision trees, votes the answer

PROS Robust · feature imp.

CONS Slower to train

Model comparison — real numbers



FULL METRICS

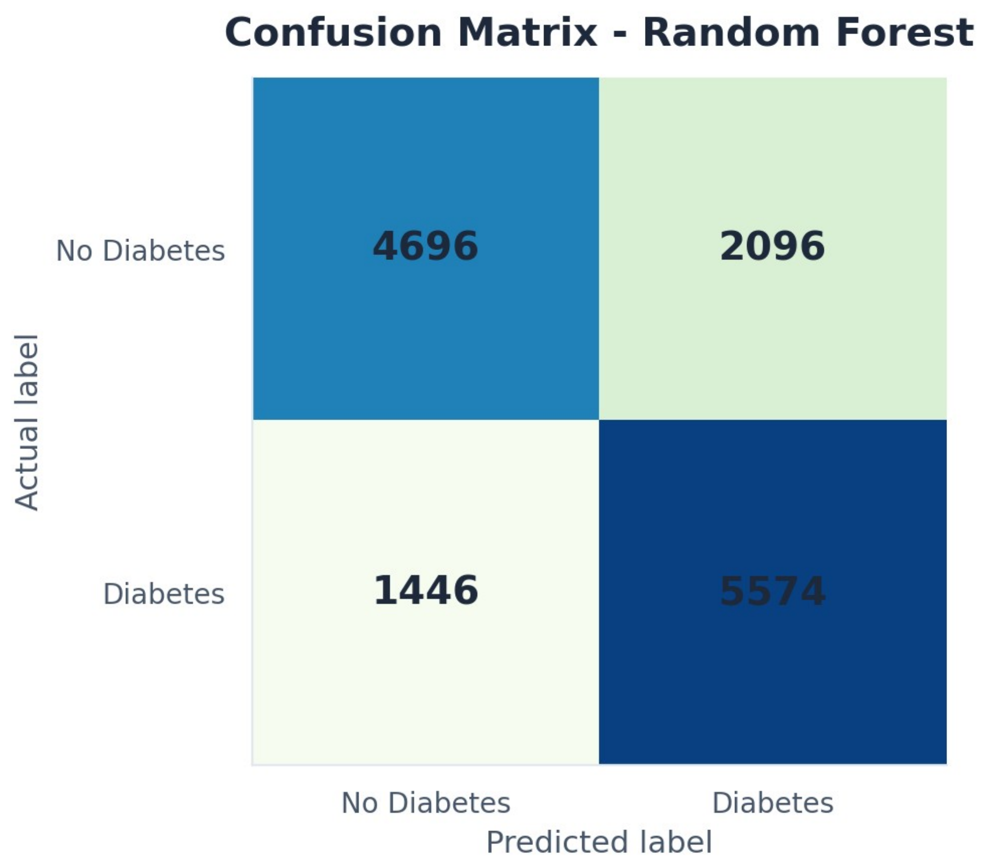
Model	Acc	F1
Logistic Regression	0.745	0.755
Decision Tree	0.732	0.747
K-Nearest Neighbors	0.735	0.752
Random Forest	0.744	0.759

WINNER

Random Forest

Best F1 (0.759) and built-in feature importance.

Random Forest — detailed results



ACCURACY

74.36%

F1 SCORE

0.759

PRECISION

0.727

Predicted diabetics that were correct

RECALL

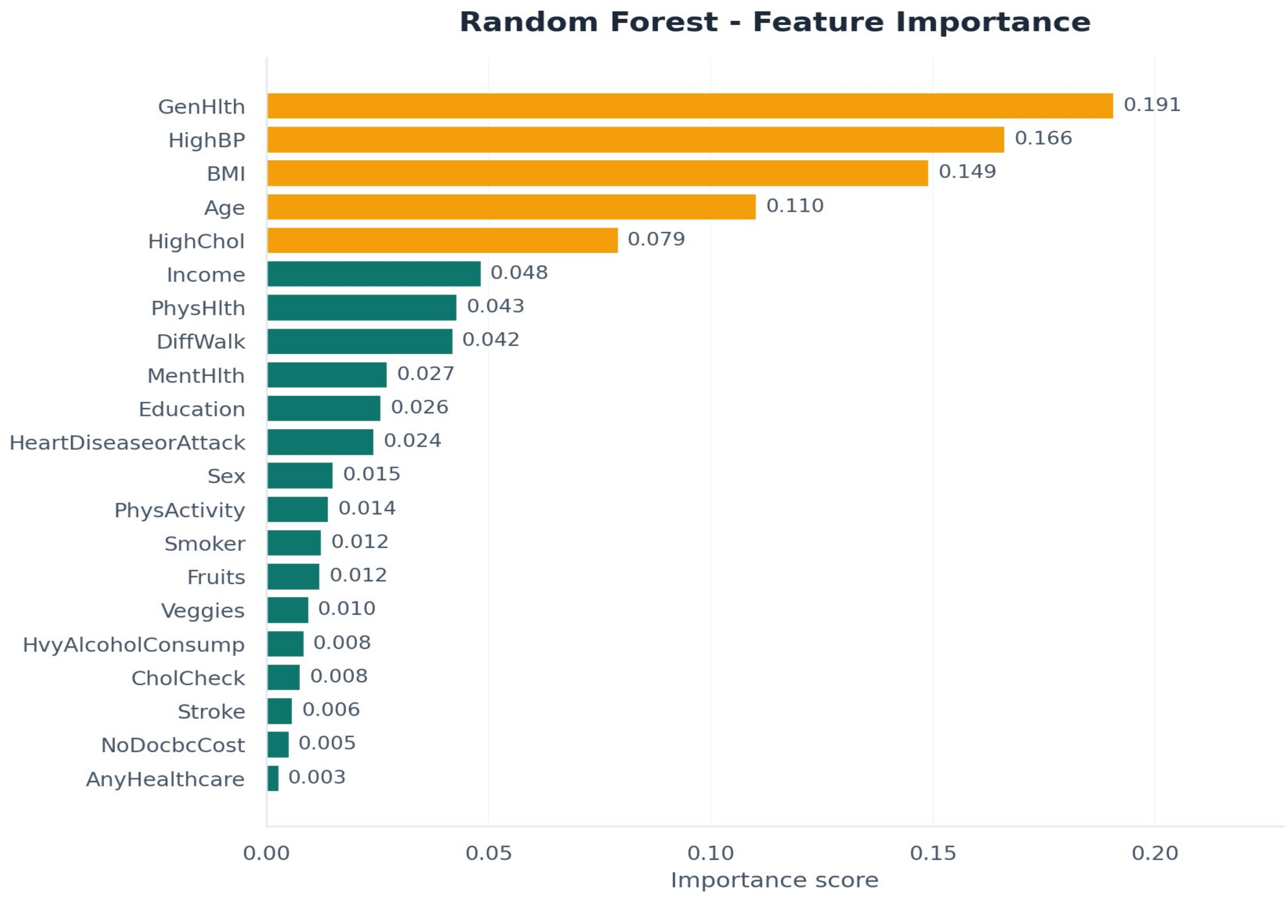
0.794

Actual diabetics we correctly caught

HYPERPARAMETERS

n_estimators = 200 · max_depth = 15 · min_samples_leaf = 5
random_state = 42 · n_jobs = -1 (use all CPU cores)

Top diabetes risk factors



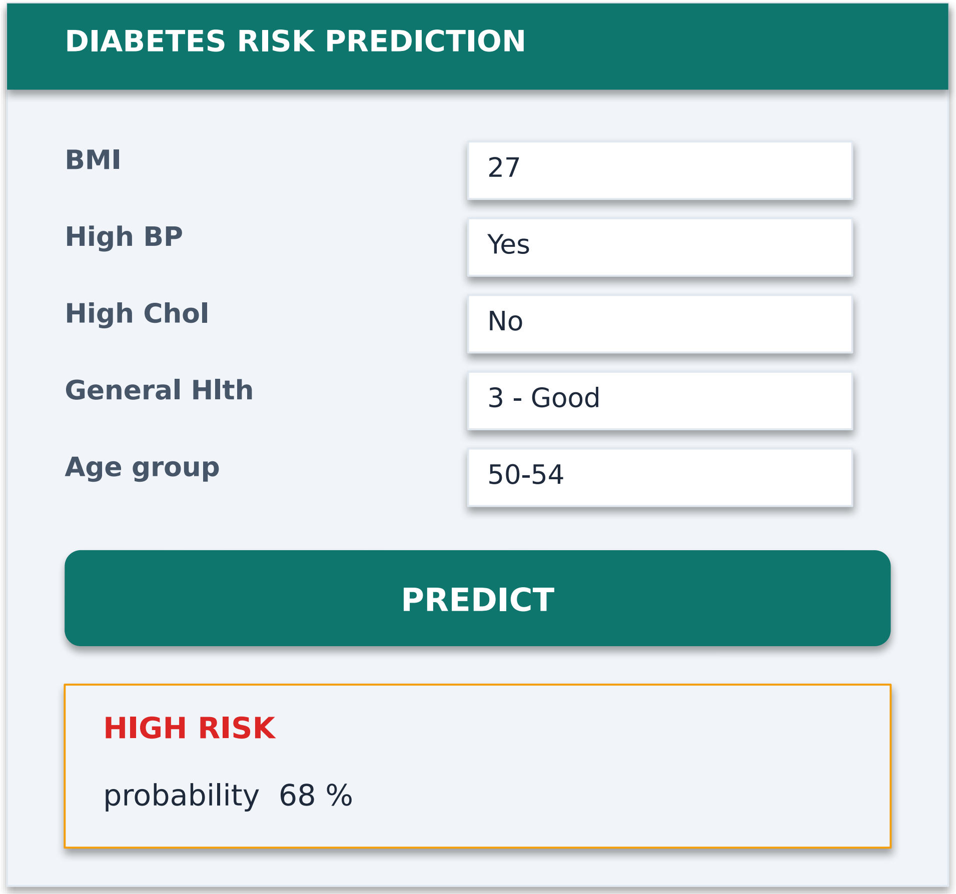
Top 5 highlighted in amber: GenHlth (0.191), HighBP (0.166), BMI (0.149), Age (0.110), HighChol (0.079)

TOP 5 RISK FACTORS LEARNED

1. **GenHlth**
   importance = 0.191
2. **HighBP**
   importance = 0.166
3. **BMI**
   importance = 0.149
4. **Age**
   importance = 0.110
5. **HighChol**
   importance = 0.079

Streamlit web application

- Interactive form built with Streamlit (Python web framework).
- User enters 21 health indicators in plain English.
- Trained model loaded once and cached for speed.
- Output: risk label + probability + confidence bar.
- Code on GitHub · deployed free on Streamlit Cloud.



Limitations · Conclusion · Q&A

LIMITATIONS

- Self-reported survey data is noisy.
- No lab values (no HbA1c or glucose).
- Balanced data ≠ real-world prevalence (~14%).
- Class 1 mixes diabetic + prediabetic.

CONCLUSION

- Random Forest classifier · 74.4% test accuracy · 0.759 F1 score.
- Identified 5 dominant risk factors aligned with medical literature.
- Delivered a working web app and reproducible notebook.

LIKELY QUESTIONS

Why Random Forest over Logistic Reg?

Best F1 + feature importance for free.

Why only 74% accuracy?

Self-reported data caps it; matches published research.

Why drop duplicates?

Prevents test-set inflation (data leakage).

Supervised or unsupervised?

Supervised - target is labeled 0 / 1.

How would you improve it?

Add lab values · try XGBoost · tune hyperparameters.

Thank you · Questions?